

Instructor: Date:

Learner: Regional Branch:

Program: Center:

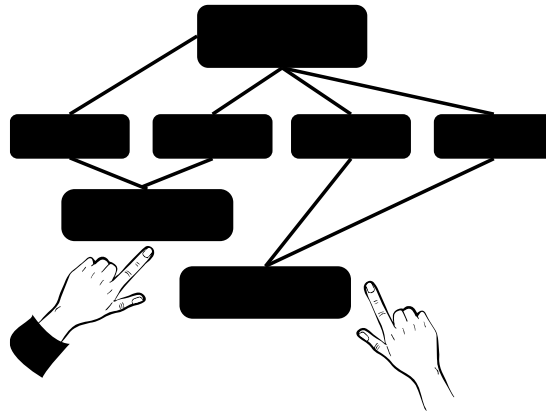
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Let's Start!

A Standard Language for Designing



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.

Explain important features related to object programming in different contexts and with different types of object programming.

Can Do Statements:

- I can explain what UML is.
- I can understand why UML plays an important role in software programming.
- I can give examples of some of diagrams used in UML.

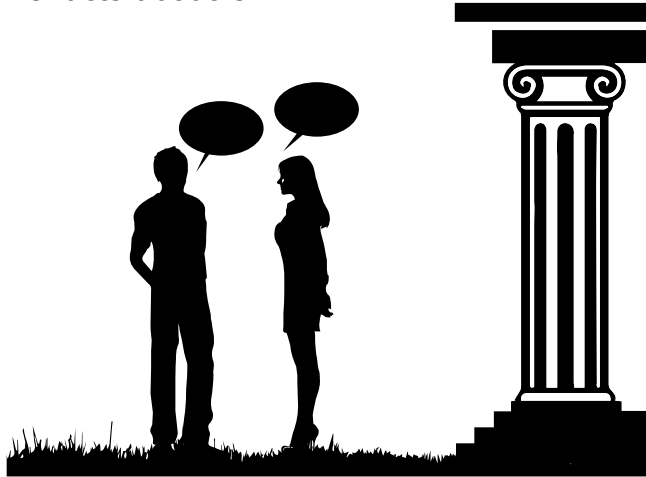
Ask your partner:

- Why is it important to design programs before coding them?
- What are the benefits of using a modeling language when designing a software?



Let's Explore!

Tara explains some facts about UML.



Instructions: Read the text below.

Erik: Hey Tara, I need your help. I need to prepare a presentation about the unified modeling language for my programming class and I know you are an expert. Do you have time to help me out?

Tara: Definitely! That's one of my favorite topics. You need to start your presentation by explaining what UML is.

Erik: Right. But, I don't understand the concept very well.

Tara: It's very easy. You can start by saying that the unified modeling language, also known as UML, is a useful and standard language that programmers use to design and document a system. UML uses an object-oriented style in software projects.

Erik: OK. Does UML play an important role in software programming?

Tara: Of course. This model describes software like the different functions of a system from the user's perspective, the systems' attributes, relationships, and objects. It's important to mention that UML doesn't need the software's coding because it is not necessary to show the design of a software architecture.

Erik: Great! So, does UML have any kind of established procedures? I saw some big, rectangular, colorless charts on one of my books.

Tara: Well, there are some diagrams that people use in UML. For example, Activity which shows the data flow and processes in business transactions. There are others like Class where you can see the types of data; Communication where classes exchange messages; Packages where you organize all the elements of the object; Component where the major parts of the system appear. Oh! I almost forgot. Deployment which is the one that deploys the systems including hardware and software servers.

Erik: Thanks a lot, Tara! You are very smart.



**Let's Practice!**

Tara explains what UML is.

Instructions: Organize the mystery sentence.

1. language / document / standard / UML / a / is / useful, / to / a / system.

2. some / UML / rectangular / big / charts. / has

3. new / a / practical, / is / UML / way / a / design / software. / to

**Let's Practice!**

Why should we use UML for designing software?

Instructions: Match column A with column B.

COLUMN A

1. UML describes functions
2. UML shows the software
3. UML does not require

COLUMN B

- a. software's coding only conveys relationships and objects.
- b. of the system from the user's perspective.
- c. architecture and the attributes of the system.



Extension Activity

Let's learn a little bit more about UML.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for examples about more UML diagrams.
- Explain the diagrams you find.
- Identify their main characteristics.
- Provide examples of situations in which you would implement a diagram.
- Share your findings with a partner.

Self-Assessment

Yes

No

- I research about UML diagrams.
- I explain the features of UML diagrams.
- I identify the main characteristics of the diagrams I found.
- I think about possible situations in which I would implement a diagram.
- I read my findings with a partner.

☐	☐
☐	☐
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Target Words

- **model** (noun): a representation of something
e.g. This model describes software like the different functions of a system from the user's perspective, the systems' attributes, relationships, and objects.
- **activity** (noun): processes and data flow that occur in a transaction
e.g. Activity which shows the data flow and processes in business transactions.
- **class** (noun): command that shows the type and relationship of components
e.g. There are others like Class where you can see the types of data.
- **component** (noun): important part of a system
e.g. Components where the major parts of the system appear.
- **deploy** (verb): to organize objects strategically
e.g. Deployment which is the one that deploys the systems including hardware and software servers.
- **package** (noun): elements organized in to a group
e.g. Packages where you organize all the elements of the object.



Learning Points

The Simple Present Tense Review

You use **the simple present tense** to talk about routines and daily activities.

It's time to review the simple present tense in all its forms (affirmative, negative, and questions).

Affirmative: When we use present simple in affirmative sentences we just need to be careful when conjugating the verbs since the verbs are different for the subjects in third person.

Structure:

Subject + **verb** + complement

Examples:

*I **use** the unified modeling language to design what I need.*

*He **uses** the unified modeling language to design what I need.*

Negative: To present negative statements in simple present, we just need to use the auxiliary do not (don't) and does not (doesn't) depending on the subjects of the sentence.

Structure:

Subject + **do not (don't) / does (doesn't)** + verb in base form + complement

Examples:

*A Programmer **doesn't** code the software without a previous design.*

*Programmers **don't** code the software without a previous design.*

Questions: To ask questions using present simple you just need to add the auxiliary do or does (depending on the subject of the sentence) at the beginning of the statement.

Structure:

Do / Does + subject + verb in base form + complement?

Examples:

Do programmers need a modeling language?

Does a programmer need a modeling language?

Examples

- *Does UML play an important role in software programming?*
- *Does UML have any kind of established procedures?*
- *You need to start your presentation by explaining what UML is.*
- *UML uses an object-oriented style in software projects.*
- *I don't understand the concept very well.*





Learning Points

Order of Adjectives

In English, there is a specific **order for adjectives** when you need to mention more than one for an object

This is the order in which adjectives should appear in case that you need to provide a complete description of something.

Order	Description	Example
1	Opinion	A useful software
2	Size	A big tool
3	Physical quality	A rough description
4	Shape	A square system
5	Age	An old procedure
6	Color	A red light
7	Origin	A German software
8	Material	A plastic device
9	Purpose	A programming device

Let's see some examples of nouns with more than one adjective:

*I need a **practical**, **light**, and **new** software for the company.*

Examples

- *The unified modeling language, also known as UML, is a useful and standard language.*
- *I saw some big, rectangular, colorless charts on one of my books.*